➕ Convolution: The Distribution of a Sum

The goal of convolution is to find the probability distribution function ($P_Z(z)$ or $f_Z(z)$) for a new random variable Z that is defined as the sum of two independent random variables, X and Y.

The core idea is to account for **all possible pairs of (x, y) that sum up to** z and add (or integrate) their probabilities.

1. The Intuition Behind Convolution

Imagine you want to find the probability that $Z = 5$. Since $Z = X + Y$, the possible ways this can happen are:

- $X = 0$ and $Y = 5$
- $X = 1$ and $Y = 4$
- $X = 2$ and $Y = 3$
- ...and so on.

Because X and Y are **independent**, the probability of each specific pair occurring is $P(X = x) \cdot P(Y = y)$. To get the total probability $P(Z = 5)$, you must sum the probabilities of all these mutually exclusive scenarios.

$$\text{Total Probability } P(Z = z) = \sum_{\text{all possible } x} P(X = x) \cdot P(Y = z - x)$$

2. Convolution in Discrete Distributions (PMF)

If X and Y are independent discrete random variables with PMFs P_X and P_Y, the PMF of $Z = X + Y$ is given by:

$$P_Z(z) = \sum_x P_X(x) P_Y(z - x)$$

- **The summation variable** x runs over all possible values of X.
- This is often denoted by the asterisk $\text{notation: } P_Z = P_X * P_Y$.

3. Convolution in Continuous Distributions (PDF)

If X and Y are independent continuous random variables with PDFs f_X and f_Y, the PDF of $Z = X + Y$ is given by the **convolution integral**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$$

- **The integration variable** x runs over all possible real values.
- The argument $z - x$ ensures that the product $f_X(x) f_Y(z - x)$ captures all pairs (x, y) such that $x + y = z$.
- **Alternative form:** Due to the commutative property of convolution, the formula can also be written as:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy$$

4. Key Results and Applications

You don't always have to calculate the integral/sum. Many common distributions have "closure properties" under convolution:

Original Distributions	**Sum Distribution $Z = X + Y$**	**Application**
$\text{Normal}(X_1, \sigma_1^2) + \text{Normal}(X_2, \sigma_2^2)$	**Normal** $(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$	This is why the **Central Limit Theorem** works.
$\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2)$	**Poisson** $(\lambda_1 + \lambda_2)$	The total number of events from two independent Poisson processes is also Poisson.
$\text{Binomial}(n_1, p) + \text{Binomial}(n_2, p)$	**Binomial** $(n_1 + n_2, p)$	The total number of successes in two separate sets of trials.

Do you want to proceed with the final major topic, **Order Statistics**?